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Attitude motion classification of resident space objects using light curve spectral analysis

G. Isoletta *, R. Opromolla, G. Fasano

Department of Industrial Engineering, University of Naples "Federico II", Piazzale Tecchio 80, Naples 80125, Italy

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Abstract

The characterization of Earth-orbiting objects in terms of attitude motion, material composition, and shape is crucial for Space Domain Awareness. In this respect, light curves, i.e., graphs showing the brightness variation of space objects over a specific time interval, can be used to infer both their dynamical and physical properties. However, such light curve inversion problem is challenging due to measurement noise, data gaps, and the intrinsic difficulty in separating the multiple effects impacting on the object's brightness. Existing light curve inversion methods rely on estimation filters exploiting data fusion, Machine Learning approaches, or are based on the analysis of the light-curve frequency spectra. Many of these approaches make assumptions about a-priori knowledge of target object's characteristics, such as shape, size, surface reflective properties, and/or observation parameters, e.g., target's orbit and observation geometry. In this framework, this paper presents an innovative architecture for the classification of the attitude motion of space objects using only photometric light curve data, not requiring any a-priori assumption or knowledge. The proposed architecture exploits the Lomb-Scargle Periodogram algorithm to retrieve a frequency spectrum from light curve measurements, and then performs the classification by analyzing the spectrum characteristics testing different candidate rotation periods through the Phase Dispersion Minimization method. The proposed classification algorithm is first tested using a light curve simulator, in which any complex geometric model and different surface properties can be generated. Finally, the method is applied to real light curves retrieved from a publicly available database to assess the classification performance.

Keywords: Space Situational Awareness; RSO characterization; Attitude motion classification; Light curve inversion

1. Introduction

Space Domain Awareness (SDA) is pivotal in the characterization and management of Earth's resident space objects (RSOs). A key aspect of SDA involves the extraction of crucial information such as attitude motion, material composition, and shape (Fan and Frueh, 2020; Šilha et al., 2021). Indeed, accurate knowledge of these properties enhances object tracking and identification, thereby contributing to the development of robust space traffic management (STM) systems. Within this framework, photometric information, especially in the form of light curves, has emerged as a valuable resource to infer both dynamical and physical properties of space objects (Šilha et al., 2021). A light curve is a graph containing measures of the brightness of an object over time based on reflected sunlight, as observed by an optical sensor. The observed brightness is influenced not only by intrinsic object properties (such as shape, materials, and attitude motion), but also by the observation geometry (which in turn depends on the relative positions of the object, the observer, and the Sun). The accurate determination of object properties from light

* Corresponding author.
E-mail addresses: giorgio.isoletta@unina.it (G. Isoletta), roberto.opromolla@unina.it (R. Opromolla), g.fasano@unina.it (G. Fasano).

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curves is crucial for diverse applications in space surveillance and debris mitigation (Blacketer et al., 2022; Vallverdú Cabrera et al., 2021), despite the need of dealing with noise issues arising from light contamination from other objects, atmospheric effects, and limitations in optical sensors' detection (Fan and Frueh, 2020; Kerr et al., 2021; Piergentili et al., 2017).

Light curve inversion is the reconstruction of the intrinsic properties of an object by analyzing its observed brightness variation over time. This process is a non-trivial operation due to the complex interplay among the object's properties, observation geometry, and various noise sources. Several methods have been proposed in the literature to tackle light curve inversion, which can be broadly categorized into three main classes: filtering techniques based on estimation theory; Machine Learning (ML)-based approaches; techniques involving the analysis of frequency spectra. Filtering techniques involve the combination of multiple sources of information, i.e., photometric (such as light curve measurements) and/or astrometric (such as angular measurements from optical sensors), to recursively determine the most probable shape and attitude motion of RSOs typically using non-linear Kalman Filters (KF). One first example is the Unscented Kalman Filter (UKF) proposed by Wetterer and Jah (Wetterer and Jah, 2009) to estimate the attitude and angular rates of RSOs from light curves, assuming known surface properties and shape. Later, several works have addressed the problem by applying the Particle Filter (PF) to retrieve Euler angles and attitude rates of agile objects, introducing shape model uncertainty (Holzinger et al., 2014) and improved process noise model (Coder et al., 2017). Linares et al. (Linares et al., 2014) also tested the PF approach on uncontrolled objects estimating Generalized Rodrigues Parameters (GRP). Additionally, an adaptive Gaussian mixtures UKF, proposed by Vallverdú Cabrera et al. (Vallverdú Cabrera et al., 2023), deals with attitude determination from photometric data, with the Gaussian mixture designed to capture the high non-linearities of the light curve measurement model. Then, the multiple-model adaptive estimation (MMAE) approach, consisting of a parallel bank of filters operating under different hypothesis (i.e., RSO's candidate shape), was proposed by Linares et al. (Linares et al., 2013) to enable shape estimation. More recently, Linares and Crassidis (Linares and Crassidis, 2017) proposed a variant of the PF for simultaneous attitude and shape estimation, adding a shape parametrization based on facet normals and albedo-area products into the state. Another approach to solve the attitude estimation problem using light curve measurements has been proposed by Burton and Frueh (Burton and Frueh, 2021), considering known shape, reflective properties and orbit: it uses a gating technique and a probability hypothesis density (PHD) filter to identify sets of feasible orientations for each measurement, solving the so-called "pseudo-measurement association problem"; then, multiple attitude time histories are computed based on the possible orientations using a Multiplicative Extended Kalman Filter (MEKF).

As regards the ML-based light curve inversion approaches, a genetic algorithm has been used to solve a multi-objective optimization problem for attitude reconstruction, enabling the minimization of residuals between real and simulated light curves (Piergentili et al., 2021, 2017). By iteratively adjusting the object's properties, such as shape and attitude motion, the optimization process aims to find the best-fit model that reproduces the observed light curves, leveraging the principles of evolutionary algorithms to obtain solutions that closely match the real data. Linares et al. (Linares et al., 2020) proposed a data-driven method for RSOs' light curve classification based on a deep Convolutional Neural Network (CNN). A physics-based model, reproducing RSO's reflected light as a function of size, shape and attitude motion, has been used to generate thousands of light curves employed to train the CNN, which has also been tested using real data. Another ML technique has been proposed by Badura and Valenta (Badura and Valenta, 2023) to estimate the spin properties of satellites from their light curves, using a physics-based loss function in the learning objective of the ML model. In this way, the spin state predictions made by the time-variant ML model result to be physically consistent, showing low prediction errors at the same time. Finally, Qashoa and Lee (Qashoa and Lee, 2023) proposed a wavelet scattering transformation to extract features from light curves to be provided as input for an ML classifier. The light curve classification is then performed both using a conventional ML approach, i.e., support vector machine (SVM), and a deep learning one, i.e., long short-term memory (LSTM), the latter outperforming the former for LEO light curve classification between stable satellites, tumbling satellites and rocket bodies.

The analysis of the frequency spectra obtained from light curves offers an alternative method for extracting attitude information, such as the spin periods of space debris objects. These methods focus on identifying periodicities and harmonic components in the light curves to infer the rotation or oscillation characteristics of the objects, thus providing insights into their dynamics. Commonly employed techniques for detecting rotation periods from light curves include Fast Fourier Transform (FFT) and Periodogram analysis, although they need pre-processing steps to get evenly spaced data (Linder et al., 2015). Other methods, such as Epoch Folding (Larsson, 1996) and the Lomb-Scargle Periodogram (LSP), can be applied to unevenly sampled time series data and have gained significant popularity in the field of space object characterization. Especially, LSP has been extensively used in prior studies aiming to estimate spin periods of observed objects (Linder et al., 2015), proving to be highly reliable especially in the analysis of sparse and noisy data (VanderPlas, 2018). In particular, Quint et al. (Quint et al., 2022) proposed a process in which the LSP is first computed to detect all possible candidates for the rotation frequency, and then an iterative epoch folding method is applied to each candidate to optimize the frequency estimation, using different types of SSA measurements, i.e., light curves, radar cross section and laser range measurements.

Currently, many of the reviewed approaches make assumptions about a-priori knowledge of target object's characteristics, such as shape, size, surface reflective properties, and/or observation parameters, e.g., target's orbit and observation geometry.

Instead, as a first main contribution, this paper proposes an original processing architecture for the classification of the attitude motion of observed RSOs based only on photometric light curve data, thus not relying on any a-priori information about orbit, shape and physical characteristics of the objects. The presented architecture exploits the Lomb-Scargle Periodogram algorithm to retrieve the frequency spectrum from the light curve measurements, and then performs the classification by analyzing the characteristics of the spectrum, e.g., peak frequencies and their periodicity, and using a Phase Dispersion Minimization (PDM) technique (Stellingwerf, 1978) to test the candidate rotation periods. Since the classifier function requires the appropriate selection of a set of user-defined parameters, the selection logic and its results are first discussed and justified.

As a second contribution, the classification algorithm is validated using synthetic light curves generated by means of an ad-hoc developed simulator, which allows to reproduce any complex geometric model and surface characteristics, as well as different observation geometries and attitude states. In particular, the performance dependence on spin rate and observation duration is analysed for spin-stabilized test cases, while the impact of different initial attitude states and object's shapes on classification accuracy is investigated for tumbling test cases.

As a third main contribution, the method is applied to real light curves extracted from a publicly available light curve database to further assess the performance of the attitude motion classifier.

The paper is organized as follows. In Section 2, the light curve classifier is presented, with a particular focus on the criteria exploited for the classification task. Section 3 describes the light curve simulator, analyzing both the dynamics aspects and radiometric ones like the implementation of the Bidirectional Reflectance Distribution Function (BRDF) model. In Section 4, the selection of the user-defined parameters required by the classifier is described and justified, and the results of the classification for simulated test cases are presented and discussed. Performance of the method applied to a real light curves dataset is assessed in Section 5. Finally, conclusions are drawn in Section 6.

2. Attitude motion classifier

In this Section, an architecture for the broad classification of the attitude motion of space objects using only photometric light curve data is presented. The architecture is based on a spectral analysis of the light curve, and it can distinguish between three possible classes: "Spin-stabilized", i.e., objects rotating with a stable angular velocity around a principal axis; "Tumbling", i.e., objects undergoing uncontrolled rotation about multiple axes, without a stable principal axis; or "Uncertain", when the data are deemed not reliable enough to confirm either a spin-stabilized or tumbling state. It is worth noting that this classification can be seen as an extension of the light curve's periodic behaviour analysis, meaning that, for example, a light curve exhibiting a periodic behaviour is more likely to correspond to a spin-stabilized attitude motion, and thus this classification is assigned.

Among the possible options for spectral analysis, the Lomb-Scargle Periodogram (VanderPlas, 2018) was preferred for its capability of handling data unevenly spaced or with gaps, which is likely when dealing with real observations performed by ground-based telescopes. In practice, the LSP fits sinusoidal functions to the observed data at various frequencies, and then, using a least-squares approach, it quantifies the significance of each frequency component over a specific range (defined based on the signal sampling time) by calculating the resulting power spectrum. Higher power values indicate stronger periodic signals, allowing for the identification of rotation or oscillation periods in the light curves (Lomb, 1976; Scargle, 1982; VanderPlas, 2018). In addition, the PDM algorithm has been implemented and used to test the candidate rotation periods identified by the LSP. The workflow of the classification algorithm is shown in Fig. 1, where the specific conditions "Spin-stabilized frequency check" and "Frequency validation check" to be verified vary depending on the followed branch of the workflow, i.e., on the number of detected peaks in the periodogram passing the filters.

The proposed architecture is independent from previous knowledge about the object, such as details about its geometry, or from previous observations. The main processing blocks shown in Fig. 1 are described in detail below.

First, a pre-processing step, called detrending, is applied to the input light curve. Detrending is typically needed, especially for LEO objects, since it allows removing a trend in the object's brightness as captured by an optical sensor due to rapid changes of the observation geometry (as they are assumed to be unrelated to the attitude motion). Detrending methods can be physics-based or analytical-based. In the former case, the light curve corresponding to a Lambertian sphere in the same conditions of the object of interest is computed and subtracted from the actual light curve. In the latter case, the curve to subtract is evaluated by fitting the data using a polynomial function (Linder et al., 2015). In the proposed architecture, an analytical approach is adopted using a quadratic polynomial fitting function (Quint et al., 2022). It is worth noting that visual magnitude calibration and range-normalization steps, which represent typical pre-processing steps when dealing with raw light curves, are not required in this work. Indeed,

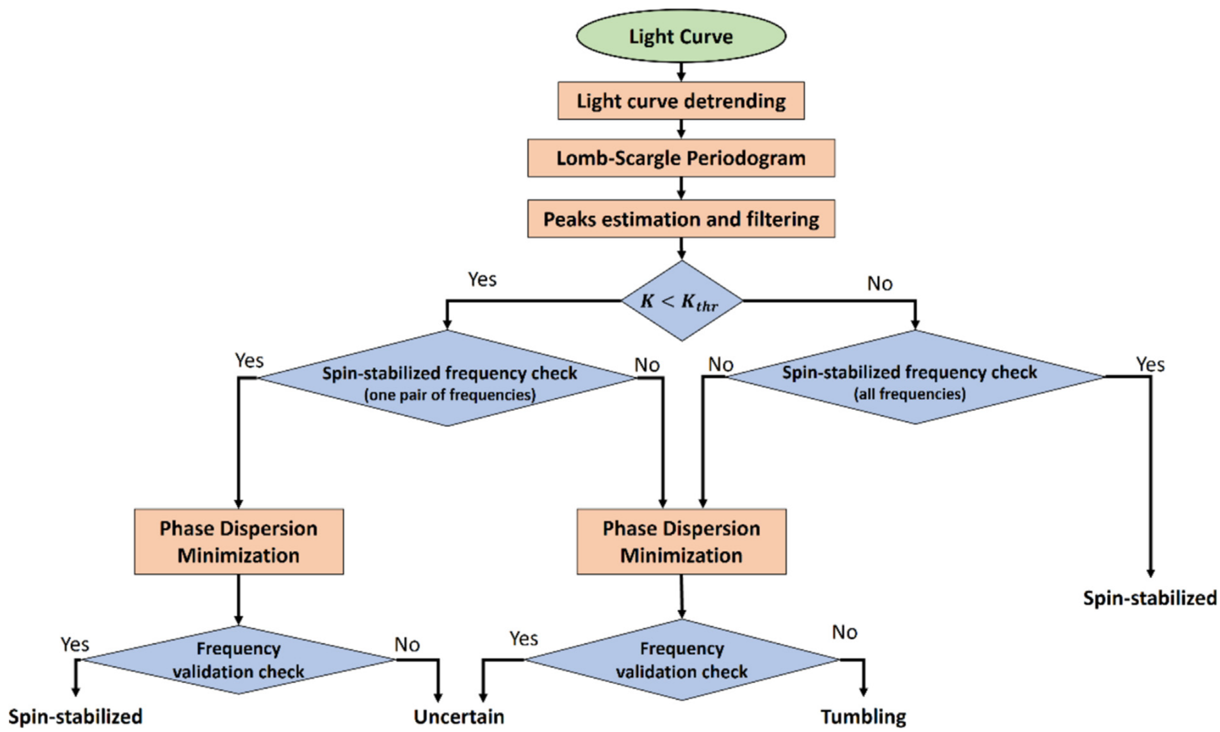

Fig. 1. Workflow of the classification algorithm. Here, K is the number of peaks passing the filter and K_{thr} is a threshold on the number of peaks to discriminate between the two branches of the algorithm.

the synthetic light curves are generated based on direct computation of the apparent visual magnitude, accounting for the object's geometry, reflective properties, illumination, and attitude motion. while the real observational data have already been calibrated before being published.

Then, the LSP is computed starting from the detrended light curve, $g(t)$. Given n discrete samples of the light curve at times t_i (g_i), the LSP is defined as follows,

$$P_{LS}(f) = \frac{1}{2}\left\{\frac{\left[\sum_{i=1}^{n} g_i cos(2\pi f(t_i - \tau))\right]^2}{\sum_{i=1}^{n} cos^2(2\pi f(t_i - \tau))} + \frac{\left[\sum_{i=1}^{n} g_i sin(2\pi f(t_i - \tau))\right]^2}{\sum_{i=1}^{n} sin^2(2\pi f(t_i - \tau))}\right\} \quad (1)$$

where τ is specified for each frequency f as shown below.

$$\tau = \frac{1}{4\pi f} tan^{-1}\left(\frac{\sum_{i=1}^{n} \sin(4\pi f t_i)}{\sum_{i=1}^{n} \cos(4\pi f t_i)}\right) \quad (2)$$

A high value of P_{LS} at a frequency f indicates a strong component of the collected signal around a period $T = 2\pi/f$. Although the Nyquist limit is not properly defined for unevenly spaced data (VanderPlas, 2018), the periodogram is here computed until a Nyquist-like limit is reached, i.e., up to $f = f_{Ny} = \frac{1}{2\Delta t}$, where Δt is the sample spacing of the input data. Indeed, outside the interval defined by the following relation, i.e., $0 < f < f_{Ny}$, the periodogram is a series of perfect aliases of the spectrum within this range.

After the computation of P_{LS}, the peaks in the frequency spectrum are estimated and filtered. A peak is identified when two conditions on a "minimum height" and a "minimum prominence" are simultaneously satisfied. Details on the definition of the prominence associated to a peak and on its computation are available in (The MathWorks Inc, 2023).

First, all the P_{LS} values higher than the minimum height threshold, which is defined as a fraction (h) of the maximum value in the periodogram ($P_{LS_{max}}$), are detected. The resulting set of peaks is then filtered discarding the P_{LS} values whose prominence is lower than a minimum threshold, which is defined as a fraction (p) of the maximum prominence value in the periodogram (Pr_{max}). Both h and p are user-defined parameters whose setting is discussed and justified in Section 4.

At the end of this peak estimation process, an additional filtering step is required to discard potential outliers or redundant values. For instance, two close peaks are considered as representative of the same frequency, thus they need to be merged. In particular, two peaks are combined in a single one, i.e., the highest between the two, if their distance in the frequency domain is lower than a threshold, Δf_{thr}, which represents a difference in frequency from the highest one corresponding to a variation of 0.1 s in the rotation period. So, a different threshold is computed for each couple of frequency peak: for instance, given two identified peak frequencies f_A and f_B, with $f_B > f_A$, the threshold is computed as follows

$$\Delta f_{thr} = \frac{0.1 f_A^{\,2}}{1 - 0.1 f_A}, \quad (3)$$

and the two peaks are merged in a single one if $f_B - f_A < \Delta f_{thr}$. This means that the value of Δf_{thr} poses a limit on the desired accuracy in the estimation of the main period characterizing the attitude motion.

Very low-frequency peaks which arise from the finite observation window (i.e., up to two times $\frac{1}{T_{obs}}$, being T_{obs} the time length of observation of the light curve under analysis), are also not expected to be representative of the actual rotation of the object and can thus be neglected.

At this stage, the number of confirmed peaks is used as a key information to carry out the classification process. Two different pipelines are, in fact, applied depending on whether the number of peaks (K) is over or below a threshold (K_{thr}) to discriminate between the two branches of the algorithm.

If $K < K_{thr}$, the possibility to assign a "spin-stabilized" classification is investigated (i.e., the "Spin-stabilized frequency check" in the left branch of the workflow is passed) if at least two peak frequencies f_1, f_2 are found to be in a ratio of 2 or 4 (accounting for a $\pm$ 0.05 margin required for frequency discretization in LSP), as expressed by the following condition,

$$\exists f_1, f_2 : \frac{f_1}{f_2} \in [1.95, 2.05] \vee [3.95, 4.05]. \tag{4}$$

Indeed, spectral analyses conducted on spin-stabilized objects, being typically characterized by a symmetric shape, have shown frequency components around twice and/or four times their actual rotation frequency (Linder et al., 2015). If condition (4) is not met, the possibility to assign a "tumbling" classification is instead investigated. In both the cases explored depending on the result of condition (4), the idea is that the tentative classification is confirmed whether one of the selected candidate frequencies is validated by applying the PDM, as explained in detail in Subsection 2.1. If the tentative classification is "Spin-stabilized", a set of $(N_{CP} - 2)$ candidate periods uniformly sampled in the $\left(3s, \frac{T_{obs}}{2}\right)$ time interval, is tested against the periods corresponding to the peak frequencies found by LSP, i.e., $T_1 = \frac{2\pi}{f_1}$ and $T_2 = \frac{2\pi}{f_2}$, thus considering a total number of N_{CP} candidate periods. For each of these periods, the PDM analysis outputs a periodicity score (ρ): the "spin-stabilized" status is confirmed if either $\rho(T_1)$ or $\rho(T_2)$ is among the M minimum values assumed by ρ, within a time tolerance, expressed in seconds and defined as

$$\varepsilon = \max\left(\Delta t_{CP}, 1s\right) \tag{5}$$

with Δt_{CP} being the constant time difference between two consecutive candidate periods.

Instead, if the "Spin-stabilized frequency check" is not verified, and so the tentative classification is "Tumbling", the PDM is applied considering both the K peak frequencies identified by the LSP and N_{CP} candidate periods uniformly sampled in the $\left(3s, \frac{T_{obs}}{2}\right)$ time interval. In this case, the "Frequency validation check" is passed if the minimum value of ρ among those corresponding to the K LSP-based peaks (ρ_{min}) is lower than that assumed by all the N_{CP} candidate periods, and also lower than a threshold $\bar{\rho}$. Indeed, this result may be representative of an underneath spin-stabilized status, thus the object is classified as "Uncertain". Otherwise, the "Tumbling" classification is confirmed.

Clearly, N_{CP} and M are user-defined parameters, which can be set depending on the confidence level required for a correct spin-stabilized classification, as discussed in Section 4.1.

If $K \geq K_{thr}$, a "Spin-stabilized" classification is directly assigned if all the peaks correspond to integer multiples of the same frequency (again this check is done considering a 0.05 margin). Otherwise, if the "Spin-stabilized frequency check" is not verified, the PDM is applied as done for the other branch of the workflow, as evident from Fig. 1, and the same "Frequency validation check" is performed, thus classifying the object either as "Uncertain" or "Tumbling" depending on its result.

Finally, it is worth noting that, for both LSP and PDM analysis, the rotation period found is often half the true rotation period of the object (Linder et al., 2015). Therefore, once a period $\bar{T}$ is estimated from the procedure described above, the magnitude of the angular velocity is determined as $\omega = \frac{360^{\circ}}{2\bar{T}}$, expressed in degrees per second.

2.1. Phase dispersion minimization algorithm

The PDM technique (Stellingwerf, 1978) consists in folding a set of data in multiple sub-series of time length equal to a guess time interval, to verify whether a periodic behaviour emerges. The PDM is common in this kind of analysis (Larsson, 1996), and it is employed in the proposed architecture to support decision among candidates' frequencies retrieved by the LSP.

As anticipated, a given number of candidate periods, N_{CP}, is uniformly sampled from the time interval $\left(3s, \frac{T_{obs}}{2}\right)$. The lower limit of this interval (i.e., 3 s) is set since rotations with such a fast period are considered unlikely: so, folding the curve around periods too close to the sampling time may lead to errors. The upper limit, i.e., $\frac{T_{obs}}{2}$, is required to guarantee that the curve is folded at least 2 times within the observation window.

First, from the discrete set of n observations g_i, the sample variance of the whole set, σ^2, is calculated. Then, given a candidate period T^*, a phase diagram of the light curve (or "folded" light curve) is obtained by applying the modulus operator to the time axis, as in Eq. (6).

$$\Phi = \frac{\mathrm{mod}(t_i, T^*)}{T^*} \tag{6}$$

The resulting (0,1) phase interval has to be uniformly discretized into N_b bins to calculate the signal variances (s^2) over each bin, with N_b being a tuning parameter (see discussion in Section 4.1). Thus, N_b samples are obtained, each one containing n_k data points, whose variances ${s_k}^2$ can be computed. The overall variance (s^2_{tot}) for all the samples is finally obtained as reported below.

$$s_{tot}^2 = \frac{\sum_{k=1}^{N_b}(n_k - 1){s_k}^2}{\sum_{k=1}^{N_b} n_k - N_b} \quad (7)$$

Hence, the periodicity score ρ can be computed as

$$\rho = \frac{s_{tot}^2}{\sigma^2}. \quad (8)$$

From these equations it is possible to state that the periodicity score ρ is approximately equal to 1 if the candidate period T^* is not a true period, while the most likely periods are the ones which minimize ρ.

3. Light curve simulator

In this study, a simulator was developed in MATLAB environment to generate synthetic light curves of an object given its geometrical and optical properties as well as its true orbital and rotational dynamics (as obtained by proper numerical propagators) and selecting a ground- or space-based telescope. The object of interest is modelled as a combination of simple elementary shapes (sphere, cone, cylinder, parallelepiped, paraboloid), whose surfaces are sampled by means of a triangular mesh or importing any complex CAD model. With regards to the object geometrical properties, each one of the N triangular facets of the resultant model is characterized by an area (A) and an outer normal unit vector ($\widehat{n}$). Concerning the optical properties, instead, each facet is identified by a set of parameters describing its reflective properties according to the Cook-Torrance (Cook and Torrance, 1982) model. The rotational dynamics of the object is numerically propagated by integrating the torque-free Euler's rotation equations, given the inertia matrix ($\boldsymbol{I}$) and the initial attitude state, thus obtaining the rotational states over time. The propagator uses a quaternion-based attitude representation. The Euler's equations are solved using a matrix formulation, where the skew-symmetric matrix of the angular velocity is used to account for the rotational rates. The orbital dynamics, instead, is propagated using a numerical propagator which solves the equation of motion in Cowell's formulation by using a Runge-Kutta scheme of 8th-9th order (Isoletta, 2023). The propagator includes the main orbital perturbations, and the following are considered for the applications presented in this work: 20 × 20 harmonics of the EGM2008 gravity model (Pavlis et al., 2012), the NRLMSISE-00 atmospheric model (Picone et al., 2002), Sun and Moon as third-body perturbation and a cylindrical model for SRP (Vallado and McClain, 2013). It should be noted that, for the sake of simplicity, the rotational dynamics is considered to be fully decoupled from the orbital propagation, given the relatively short duration of the observation intervals.

The synthetic light curve is generated by summing the contributions of all the illuminated and visible facets, considering only the times in which the target is observable by the selected telescope. The proposed simulator can be used to replicate observations by both ground-based sensors, given their geodetic coordinates, and space-based sensors, propagating their states with the same numerical propagator. However, for the analysis shown in this paper, only observations taken by ground-based telescopes are considered. It is also worth specifying that, within the simulation environment, a 313 sequence of Euler angles (i.e., psi, ψ, theta, θ, phi, φ) is employed for the attitude parametrization, representing the rotation matrix aligning the Body Reference Frame (BRF) to the inertial frame.

A light curve is the time variation of an object's photometric brightness due to reflected sunlight, and it can be measured by the apparent magnitude (M), which is given by

$$M = -26.7 - 2.5log_{10}\left(\frac{B}{B_{Sun}}\right) \quad (9)$$

being -26.7 the apparent magnitude of the Sun, while B and B_{Sun} are the observed brightness, i.e., the irradiance, measured in W/m^2, of the target and the Sun respectively. At each time instant, the target-Sun direction, $\widehat{R_{SUN}}$, and the target-observer direction, $\widehat{R_{OBS}}$, are known. For the sake of clarity, when no further indication is provided, the unit vectors are expressed in the BRF. Otherwise, when the vectors are expressed in the Earth-Centred Inertial (ECI) reference frame, the corresponding subscript is specified.

Thus, the illuminated ($\widehat{R_{SUN}} \cdot \widehat{n} > 0$) and visible ($\widehat{R_{OBS}} \cdot \widehat{n} > 0$) facets contribute to the amount of brightness B incoming at the observer, based on the relation below

$$\frac{B}{B_{Sun}} = \sum_{j=1}^{N} A_j\left(\widehat{R_{SUN}} \cdot \widehat{n}_j\right)\frac{BRDF_j}{r^2} + B_{noise} \quad (10)$$

being r the target-observer distance, and $BRDF_j$ the Bidirectional Reflectance Distribution Function, i.e., the ratio of the reflected radiance to the incident irradiance of the j-th facet. A zero-mean white gaussian noise, B_{noise}, has been added with standard deviation (σ_{noise}) equal to μ_B/SNR, being μ_B the mean of the observed brightness and SNR a fixed signal-to-noise ratio. The SNR has been set to 20, which is considered a realistic value according to (Fan and Frueh, 2020).

To compute the BRDF, the Cook-Torrance model (Cook and Torrance, 1982) has been implemented. It has been commonly used in recent years in the SSA framework (Clark et al., 2022; Wetterer and Jah, 2009), as it is likely to represent typical spacecraft materials more accurately than other optical models (Bradley and Axelrad, 2014). According to this model, each facet contributes with a diffuse bidirectional reflectance R_d and a specular bidirectional reflectance R_s, based on their fractions d and s respectively (being $d + s = 1$). The mathematical formulation is reported in the following

$$BRDF_{Cook-Torrance} = sR_s + dR_d \quad (11)$$

$$R_d = \frac{w}{\pi} \quad (12)$$

$$R_s = \frac{D \cdot G \cdot F}{\pi\left(\widehat{R_{OBS}} \cdot \widehat{n}\right) \cdot \left(\widehat{R_{SUN}} \cdot \widehat{n}\right)} \quad (13)$$

In Eqs. (12) and (13), w is the diffuse albedo, D the Beckmann facet slope distribution function, G the geometrical attenuation factor and F the surface reflectance of a perfectly smooth, mirrorlike surface, that are given by

$$D = \frac{1}{m^2 cos^4\alpha} e^{-\left(\frac{tan\alpha}{m}\right)^2} \quad (14)$$

$$G = min\left\{1, \frac{2\left(\widehat{R_{OBS}} \cdot \widehat{n_j}\right) \cdot \left(\widehat{R_{BIS}} \cdot \widehat{n_j}\right)}{\left(\widehat{R_{OBS}} \cdot \widehat{R_{BIS}}\right)}, \frac{2\left(\widehat{R_{SUN}} \cdot \widehat{n_j}\right) \cdot \left(\widehat{R_{BIS}} \cdot \widehat{n_j}\right)}{\left(\widehat{R_{OBS}} \cdot \widehat{R_{BIS}}\right)}\right\},$$

$$j = 1, \cdots, N \quad (15)$$

$$F = \frac{(b-c)^2}{2(b+c)^2}\left\{1 + \frac{[c(b+c)-1]^2}{[c(b-c)+1]^2}\right\} \quad (16)$$

where m is the surface roughness parameter, $\widehat{R_{BIS}}$ is the unit vector bisector of $\widehat{R_{OBS}}$ and $\widehat{R_{SUN}}$, β is the angle between $\widehat{R_{OBS}}$ and $\widehat{R_{BIS}}$ and α is the angle between $\widehat{R_{BIS}}$ and $\widehat{n}$, $c = cos(\beta)$, $b = n^2 + c^2 - 1$, being $n = (1+\sqrt{w})/(1-\sqrt{w})$. The involved directions are represented in Fig. 2. Moreover, it is worth specifying that the Cook-Torrance model includes self-shadowing effects by means of the geometrical attenuation factor G, which accounts for the shadowing and masking of one facet by another, as specified in (Cook and Torrance, 1982) and references therein.

At each time instant of the simulation, the observability of the object is confirmed if the following conditions are met.

The target must be outside Earth's shadow: thus, being R_{Earth} the mean Earth radius, it is required that

$$\|R_{SAT}|_{ECI} \times \widehat{R_{SUN}}|_{ECI}\| > R_{Earth} \quad (17)$$

being $R_{SAT}|_{ECI}$ the position vector of the satellite in ECI.

To guarantee that the sensor is not dazzled by sunlight, given a dusk angle $\delta = 4^\circ$ for ground-based sensors, the following condition must be met

$$cos^{-1}\left(\widehat{R_{SAT}}|_{ECI} \cdot \widehat{R_{SUN}}|_{ECI}\right) > 90^\circ + \delta \quad (18)$$

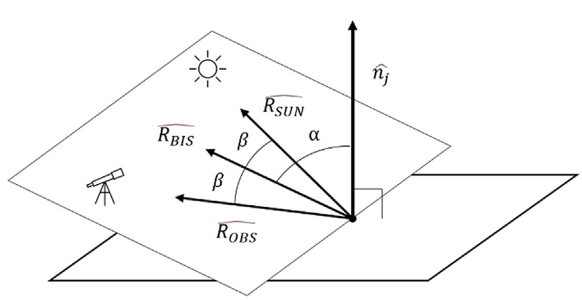

Fig. 2. Reflection geometry for the j-th facet.

while for space-based sensors with a given the field of view, FOV, (although not explicitly considered in the conducted simulations) the condition yields to

$$cos^{-1}\left(\widehat{R_{SAT-OBS}}|_{ECI} \cdot \widehat{R_{SUN}}|_{ECI}\right) > FOV \quad (19)$$

Only for ground-based sensors, the target elevation above the observer local horizon must be larger than the mask angle $\mu = 10^\circ$, thus the sensor boresight direction $\widehat{R_{OBS}}|_{ECI}$ and the observer-to-target direction $\widehat{R_{SAT-OBS}}|_{ECI} = \widehat{R_{SAT}}|_{ECI} - \widehat{R_{OBS}}|_{ECI}$ shall form an angle less than $90^\circ - \mu$. This condition can be formulated as follows.

$$cos^{-1}\left(\widehat{R_{SAT-OBS}}|_{ECI} \cdot \widehat{R_{OBS}}|_{ECI}\right) < 90^\circ - \mu \quad (20)$$

The time instants for which the conditions set by Eqs. (17), (18) and (20) are fulfilled belong to the so-called "observation window".

4. Simulated test cases results

In this section, the selection of the tuning parameters used to characterize the methodology described in Section 2 is first investigated. Then, the simulated test cases are described, and the results of the proposed attitude motion classifier are presented and discussed. Specifically, two sets of test cases are identified: for scenarios in which the target object is characterized by a spin-stabilized attitude motion, a sensitivity analysis is performed for varying object's initial attitudes and attitude rates. Instead, for scenarios in which the target object is characterized by a tumbling attitude motion the impact on performance of different object's shapes and attitude states is analysed.

Before going into the details of the choice of tuning parameters and of the analysis of results, it is worth showing an example of the effect that a different attitude motion can produce, in general, on the shape of the light curves and the corresponding periodograms. Examples of light curve and of the corresponding LS periodogram are shown in Fig. 3 and Fig. 4, respectively, using the same object model, i.e., a box-wing model satellite presented in detail in Section 4.2.2. When the target is characterized by a spin-stabilized motion, the light curve exhibits a more regular pattern with a periodic behaviour, as can also be deduced from the corresponding periodogram, showing a few number of peaks at multiple frequencies. On the contrary, the quite irregular pattern of the light curve of the tumbling case produces a very different periodogram, characterized by a high number of peaks within a narrow range of frequencies.

4.1. Selection of the tuning parameters

In the following, the motivations behind the selection of specific values for the tuning parameters are discussed.

Regarding the peak estimation and filtering step in Fig. 1, the tuning parameters, i.e., h and p, are set conduct-

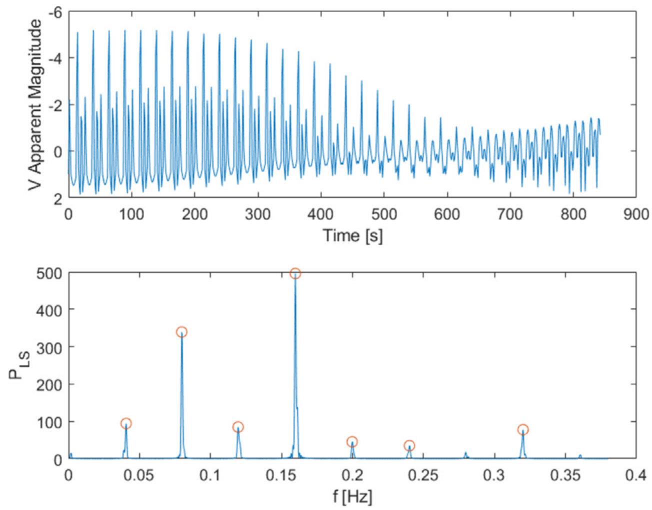

Fig. 3. Example of light curve and corresponding LS periodogram for a box-wing satellite characterized by a spin-stabilized attitude motion.

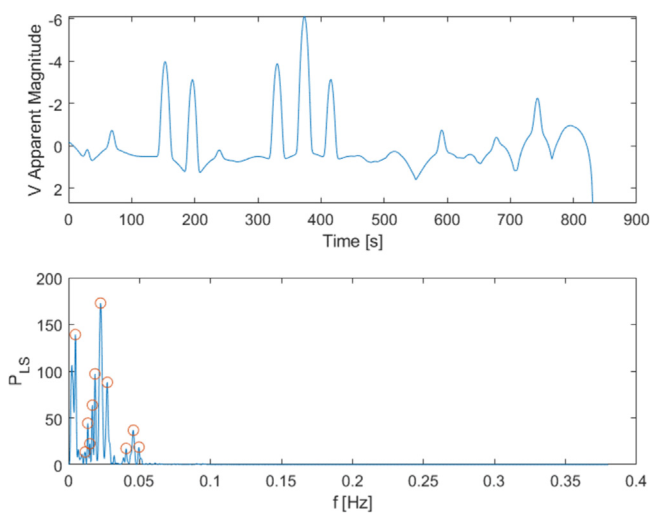

Fig. 4. Example of light curve and corresponding LS periodogram for a box-wing satellite characterized by a tumbling attitude motion.

ing a sensitivity analysis on a reference case in which four clear peaks are present in the periodogram: specifically, h and p have been set to equal values and varied from $1/20$ to $1/5$. The analysis has shown that all the peaks are identified when considering values equal to $1/15$ or smaller. Therefore, $h = 1/15$ and $p = 1/20$ have been set for the analysis presented in this section.

The threshold K_{thr} to discriminate between the "likely spin-stabilized" and "likely tumbling" pipelines has been empirically set to 7, since the periodograms associated to the light curves of tumbling objects often show a higher number of peaks with respect to those of spin-stabilized objects (Qashoa and Lee, 2023; Vallverdú Cabrera et al., 2021).

Regarding the settings of the PDM, a total number of candidate periods, N_{CP}, equal to 100 has been selected. Also, M has been set to 5 to confirm a spin-stabilized classification, and thus a periodicity around a given frequency. This guarantees that the investigated rotation period has to be between the top-5 candidate periods, i.e., the ρ value associated with that period has to be among the 5 minimum values assumed by ρ for all the 100 candidate periods. As already mentioned above, two of the 100 candidate periods are those corresponding to the two peak frequencies found by LSP, while the remaining 98 periods are uniformly sampled in the interval $\left(3s, \frac{T_{obs}}{2}\right)$. When applying the PDM to confirm a tumbling classification, instead, all the 100 candidate periods are uniformly sampled in this interval.

As for the number of bins needed to evaluate the sample variances, although the proposed methodology's performance was not found to be very sensitive to this parameter, an upper bound (i.e., $N_b < \frac{n}{5}$) is suggested for the accuracy of the method, with n being the number of light curve samples. In particular, since the PDM performs a least-squares fitting relative to a mean curve, as defined by the means of each bin, a value of N_b higher than $n/5$ could lead to a degradation of the mean curve accuracy (Stellingwerf, 1978). Therefore, N_b is selected as $n/10$, following this suggestion.

Finally, for the tumbling classification, the threshold $\bar{\rho}$ has been set to 0.7, following the observation of several spin-stabilized cases, which always show values of ρ lower than 0.6 for confirmed candidate periods.

In the following, for each simulated test case, the classification result is defined as "correct" if a spin-stabilized or a tumbling object is declared as such; instead, a missed detection is considered when the architecture produces an "uncertain" classification of a spin-stabilized or tumbling object. Finally, a wrong classification occurs if a spin-stabilized attitude motion is declared as tumbling or vice-versa. Thus, the percentages of correct and wrong classification are used as performance metrics.

4.2. Simulation results

4.2.1. Spin-stabilized scenarios

The target object model selected for the spin-stabilized scenarios is a rocket body whose geometry is inspired by that of the Atlas-Centaur II upper-stage rocket body, as in Wetterer and Jah (Wetterer and Jah, 2009). The model is composed by a 100-facet cylinder, with a radius of 1.5 m and height of 9 m, and two 100-facet hemispherical end caps extending 1 m beyond the ends of the cylinder. A representation of the model is shown in Fig. 5 and its physical and reflective properties (as defined in Section 3) are reported in Table 1.

<u>*First spin-stabilized scenario*</u>

For the first scenario, the light curve of the rocket body is simulated as seen by the Advanced Electro-Optical Sensor (AEOS) telescope of the Maui Space Surveillance Com-

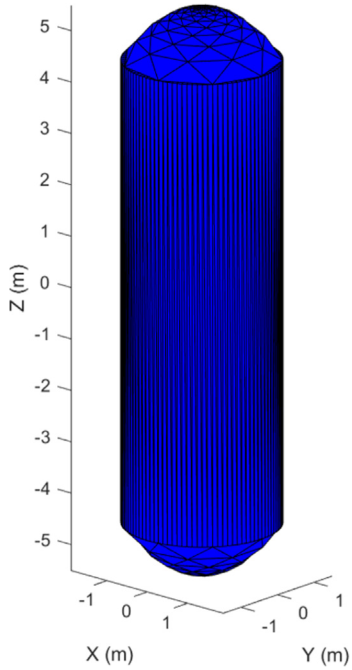

Fig. 5. Representation of the rocket body model.

Table 1
Physical and reflective properties of the rocket body model.

Parameter	Value
I_{xx}	14625 $kg \cdot m^2$
I_{yy}	14625 $kg \cdot m^2$
I_{zz}	2250 $kg \cdot m^2$
m	0.17 (both cylinder and end caps)
w	0.2 (cylinder) – 0.6 (end caps)
d	0.3 (cylinder) – 0.8 (end caps)
s	0.7 (cylinder) – 0.2 (end caps)

plex (20.708°N latitude, 156.26°W longitude, 3.075 km altitude). The initial conditions in terms of UTC epoch and orbital parameters for the numerical propagation are summarized in Table 2, in which a is the semimajor axis, e the eccentricity, i the orbit inclination, $RAAN$ the Right Ascension of the Ascending Node and AOP the Argument of Perigee. The orbit is propagated for 1 day and the first observation window has been considered for the analysis. The observation window lasts 842.3 s, from 14:47:22 to 15:01:24 of the 24th March 2003, with a measurement frequency of 1 Hz.

First, the classifier has been tested using the same initial attitude state reported in (Wetterer and Jah, 2009), i.e.,

Table 2
Initial conditions for the first spin-stabilized scenario.

UTC Epoch	*a* (km)	*e*	*i* (°)	*RAAN* (°)	*AOP* (°)
24th March 2003 13:30:00	7305.66	0.0634	30.4	77.4	323.6

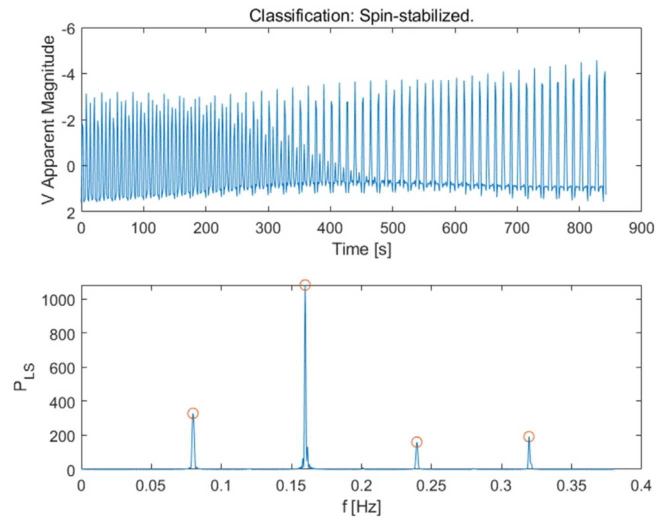

Fig. 6. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a correct "Spin-stabilized" classification for the first scenario, considering $[\psi_0, \theta_0, \varphi_0] = [11.5°, -80.2°, -17.2°]$ and $\omega_0 = [0, -14.44°/s, 0]$, without added noise.

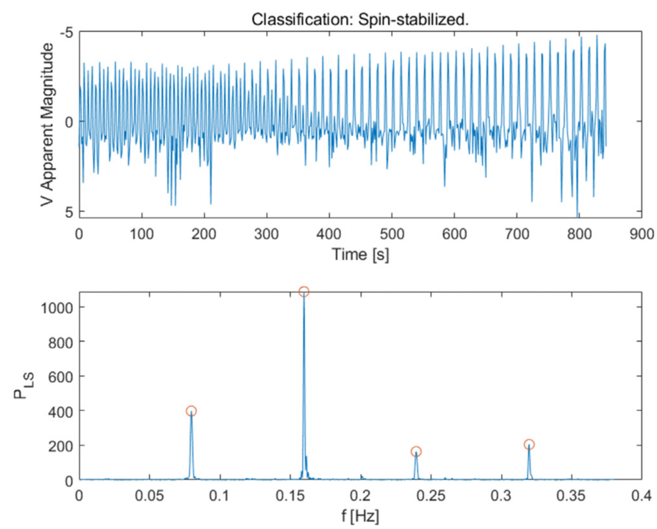

Fig. 7. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a correct "Spin-stabilized" classification for the first scenario, considering $[\psi_0, \theta_0, \varphi_0] = [11.5°, -80.2°, -17.2°]$ and $\omega_0 = [0, -14.44°/s, 0]$, with added noise.

$[\psi_0, \theta_0, \varphi_0] = [11.5°, -80.2°, -17.2°]$ as initial attitude and $\omega_0 = [0, -14.44°/s, 0]$ as initial angular velocity vector, both without and with added noise. The simulated detrended light curve and the corresponding Lomb-Scargle periodogram for both the analyses are shown in Fig. 6 (without noise) and Fig. 7 (with noise). The values of apparent magnitude of the simulated light curve after detrending are coherent with typical light curves of LEO objects (Karpov et al., 2016; "Mini-Mega-TORTORA Database," n.d.; Silha et al., 2017). Although the shape of the light curve is clearly affected by noise, the effect on the corresponding LSP is not significant, thus not altering the number nor the height of the detected peaks. In both cases, the periodogram presents a first peak at a frequency $f_1 = 0.08$ Hz and then other three peaks all at frequencies multiple of the first one. By applying the PDM, the period $T_1 = \frac{1}{f_1} = 12.5$ s shows a value of $\rho = 0.238$, being between the top-5 periods characterized by the lowest values of ρ, thus confirming the rotation period. So, the algorithm correctly classifies the attitude motion as spin-stabilized, estimating an angular velocity magnitude $\omega = \frac{360^\circ}{2T_1} = 14.4°/s$.

A sensitivity analysis has been then carried out to evaluate the limits of the spin-stabilized classification in terms of angular velocity rates and to assess the robustness of the classification algorithm against noise. A set of different spin rates has been tested for increasing angular velocity magnitude, running 100 simulations for each value of ω_0 and considering $[\psi_0, \theta_0, \varphi_0] = [11.5°, -80.2°, -17.2°]$ as initial attitude. The results of the analysis in terms of classification accuracy are reported in Table 3, for spin rates ranging from 2°/s to 60°/s. The algorithm seems to be quite robust in identifying a spin-stabilized attitude motion for spin rates ranging from 2°/s to around 30°/s. On the contrary, for spin rates higher than 30°/s, there are generally no multiple frequencies detectable in the periodogram. This misclassification as "Tumbling" is not a limitation of the classification algorithm itself, but rather a result of the sampling frequency being too low to capture the higher rotation speeds. Higher sampling frequencies would be needed to accurately resolve the rotational dynamics at these higher spin rates. It is worth noting that, in the last test case, the motion is correctly classified as spin-stabilized by chance: there are always two multiple frequencies, but they are not the "right" ones, as the wrong estimation of the angular velocity clearly proves.

Finally, another analysis has been performed to assess the dependence of the spin-stabilized classification on changes in initial attitude. To this aim, 500 spin-stabilized test cases have been simulated, keeping fixed the angular velocity to $\omega_0 = [0, -14.44°/s, 0]$. The different initial attitudes were simulated sampling the three attitude angles randomly from uniform distributions, between [-180°, 180°] for the first and third angle and between [-90°, 90°] for the second. The algorithm correctly classifies the spin-stabilized status 476 out of 500 times, tagging as "Uncertain" 20 cases and as "Tumbling" the remaining 4 cases, thus resulting in a classification accuracy of 95.2 %, 4 % of missed classification and only 0.8 % of wrong classification. An example of "Uncertain" classification is depicted in Fig. 8, in which the LS periodogram shows only one peak in the spectrum, while the light curve is characterized by a central zone with a lower oscillation amplitude with respect to initial and final regions of the observation window. This behaviour is due to the geometry of observation, where the telescope observes nearly the same faces of the cylindrical rocket-body throughout its pass. As a result, the light curve takes on an almost sinusoidal shape, with a dominant oscillation frequency and a lack of higher harmonics. This limitation in spectral complexity prevents the algorithm from detecting multiple peaks, leading to the missed classification.

Table 3
Results of the sensitivity analysis for varying spin rates for the first spin-stabilized scenario.

ω_0[°/s]	Classification performance			Estimated ω [°/s]
	Correct	Missed	Wrong	
[0, −2, 0]	100	0	0	2.03
[0, −3, 0]	100	0	0	2.94
[0, −3.5, 0]	100	0	0	3.41
[0, −5, 0]	100	0	0	4.93
[0, −7, 0]	100	0	0	6.93
[0, −14.44, 0]	100	0	0	14.35
[0, −30, 0]	100	0	0	29.98
[0, −40, 0]	0	0	100	–
[0, −50, 0]	0	0	100	–
[0, −60, 0]	100	0	0	16.59

Second spin-stabilized scenario

For the second scenario, the same rocket body model and the same telescope as the first spin-stabilized scenario have been considered, but the ephemerides have been generated starting from a different epoch. Indeed, the initial conditions for the 1-day propagation are summarized in Table 4, and the observation window lasts only 260.6 s, from 15:17:56 to 15:22:17 of 22nd April 2023, again considering a measurement frequency of 1 Hz.

A sensitivity analysis for varying initial attitude has again been carried out to study the impact of a reduced observation window on the performance. Running the simulation 500 times for different initial attitudes, the classifier

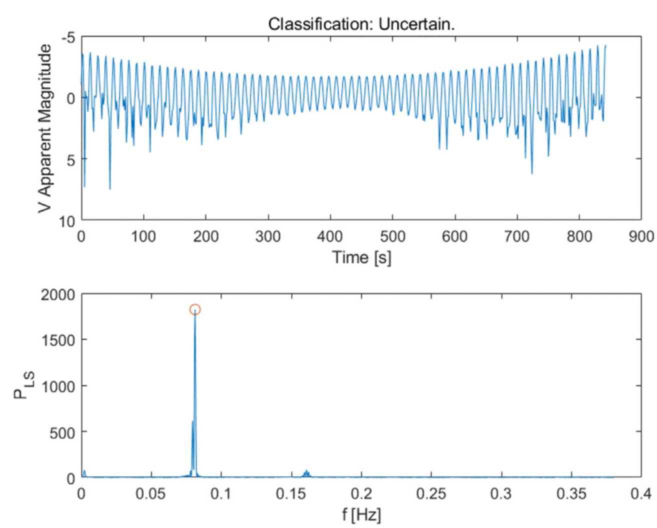

Fig. 8. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a missed "Spin-stabilized" classification for the first scenario, considering $[\psi_0, \theta_0, \varphi_0] = [30°, 85°, -87°]$ and $\omega_0 = [0, -14.44°/s, 0]$.

Table 4
Initial conditions for the second spin-stabilized scenario.

UTC Epoch	a (km)	e	i (°)	RAAN (°)	AOP (°)
22nd April 2023 00:00:58	7260.45	0.0571	30.31	330.7	142.5

has proved to have a slightly lower classification accuracy with respect to the previous scenario (87.8 %), tagging the object as "Spin-stabilized" in 439 cases, while 60 "Uncertain" classifications occur (12 % of missed classification). Only 1 case has been classified as "Tumbling", and it is depicted in Fig. 9. Here, a quite high number of non-proportional peak frequencies are identified. as the light curve presents a very irregular pattern, thus triggering the wrong tumbling classification.

An interesting analysis was performed here to better clarify the role of the detrending process. Indeed, the light curve and the periodogram of one of the "Uncertain" classification of the last analysis is reported in Fig. 10, with initial conditions $[\psi_0, \theta_0, \varphi_0] = [76°, 50°, -84°]$ and $\omega_0 = [0, -14.44°/s, 0]$. Fig. 11, instead, depicts the results of the simulation run using the same initial conditions, but without detrending the light curve given as input to the classification algorithm. It is clear that the original light curve presented illumination and/or visibility issues arising in the final part of the observation, leading to a nearly dumped trend of the apparent visual magnitude, and thus to a wrong "Tumbling" classification. Therefore, the detrending process has allowed to fix this trend and avoid the misclassification of the attitude motion. This is quite important also considering the possible operational application of this classifier, which could be used to identify potential changes or inconsistencies in the attitude motion of space objects of interest. Hence, while a wrong classification could trigger an unnecessary alert to the operators, a missed classification could be simply due to the lack of sufficient data (e.g., too short observations) or incomplete information to properly perform the attitude motion characterization, thus suggesting the operators to gather more data before repeating the analysis.

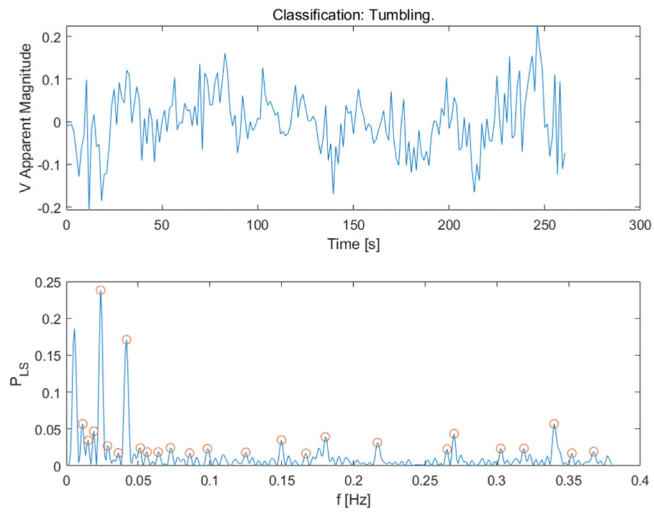

Fig. 9. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a wrong "Tumbling" classification for the second spin-stabilized scenario, considering $[\psi_0, \theta_0, \varphi_0] = [52°, 46°, 87°]$ and $\omega_0 = [0, -14.44°/s, 0]$.

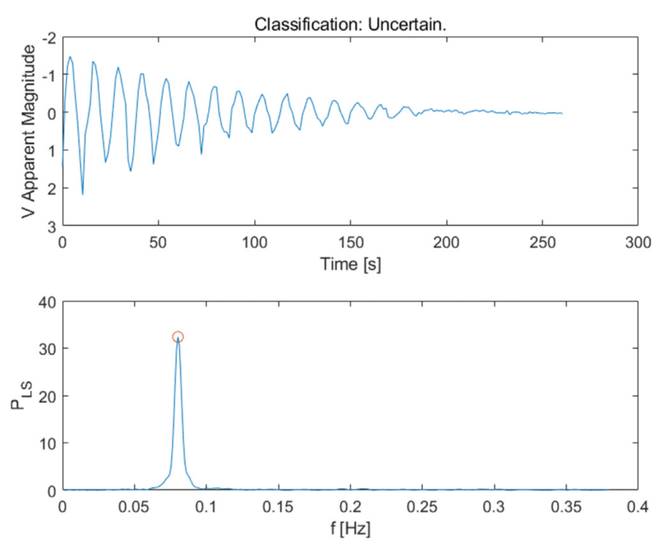

Fig. 10. Simulated light curve after detrending (top) and corresponding LSP (bottom) of an "Uncertain" classification for the second spin-stabilized scenario, considering $[\psi_0, \theta_0, \varphi_0] = [76°, 50°, -84°]$ and $\omega_0 = [0, -14.44°/s, 0]$.

4.2.2. Tumbling scenarios

For the simulated tumbling scenarios, another object's model has been built based on the Galileo's box-wing configuration ("Galileo Satellite Metadata | European GNSS Service Centre," n.d.; Montenbruck et al., 2015). The box has been modelled as a 12-facet parallelepiped of 2.5 m x 1. 2 m x 1.1 m, while the two solar panels are 1.08 m x 5 m x 0. 08 m 12-facet parallelepipeds. The roughness is assumed to be 0.17 for all the facets. The surface properties, i.e., diffuse albedo and diffuse and specular coefficients, are estimated for each facet on the basis of the information provided in ("Galileo Satellite Metadata | European GNSS Service Centre," n.d.). The representation of the box-wing model is shown in Fig. 12, while its inertia matrix, I_{BW}, is reported in Equation (21).

$$I_{BW} = \begin{bmatrix} 1230.2 & -0.3581 & -106.91 \\ -0.3581 & 724.01 & 1.8003 \\ -106.91 & 1.8003 & 1353.2 \end{bmatrix} kg \cdot m^2 \tag{21}$$

First tumbling scenario

In the first tumbling scenario, the same ephemerides, i.e., March 2003 ephemerides, and the same telescope,

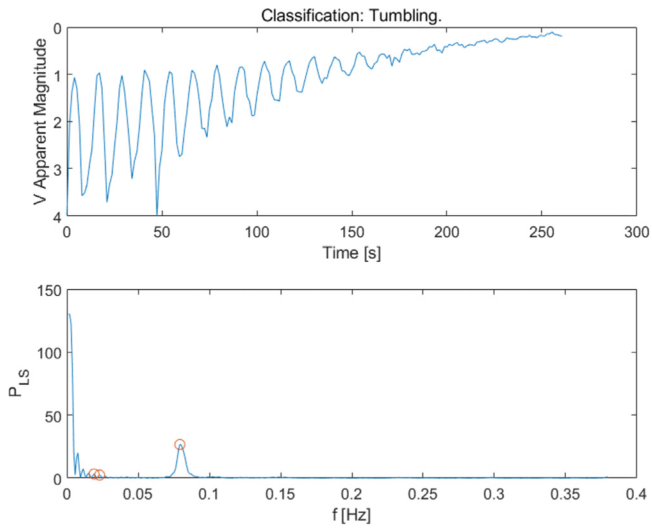

Fig. 11. Simulated light curve before detrending (top) and corresponding LSP (bottom) for the test case characterized by $[\psi_0, \theta_0, \varphi_0]$ = [76°, 50°, −84°] and ω_0 = [0, −14.44°/s, 0], for the second spin-stabilized scenario.

i.e., Maui AEOS telescope, as the first spin-stabilized scenario have been used, considering a measurement frequency of 1 Hz. Two different analyses have been run to assess the impact of different angular rates on the classification performance, and to investigate the dependence of the tumbling classification on changes in initial attitude.

In the first analysis, 500 different sets of attitude state are simulated, fixing the initial attitude to [0°, 0°, 0°] and sampling the angular velocity components from uniform distributions between 0 and 10°/s. The algorithm correctly classified as "Tumbling" 463 cases out of 500 (92.6 % of correct classification), while performing 11 wrong "Spin-stabilized" classifications (2.2 %) and 26 missed classifications (5.2 %). Two examples of a correct and a wrong classifications are shown in Fig. 13 and Fig. 14, respectively. While the irregularity of the light curve correctly classified as "Tumbling" is quite evident, as it presents several peaks grouped at the beginning of the periodogram at frequency values lower than 0.1 Hz, the light curve in Fig. 14 shows many negative peaks in the apparent visual magnitude almost evenly spaced, especially from 300 s to the end of the observation window. This is confirmed by the presence of two peaks in the LSP at frequencies one multiple of the other, which also happen to be the higher peaks in the periodogram. Both the corresponding frequencies result to be between the top-5 candidate periods after applying PDM, thus leading to the spin-stabilized classification.

Then, in the second analysis, other 500 simulations have been performed, keeping the angular velocity ω_0 fixed to [0°/s, 1°/s, 6°/s] in each simulation, and sampling the three attitude angles randomly from uniform distributions, as already done for the first spin-stabilized scenario. The classification performance resulted to be similar to the previous analysis, with 89.8 % of correct classification (449 out of 500), 3.2 % of wrong classification (16 out of 500), while tagging 35 light curves as "Uncertain". Fig. 15 presents an example of correct classification for the case with initial

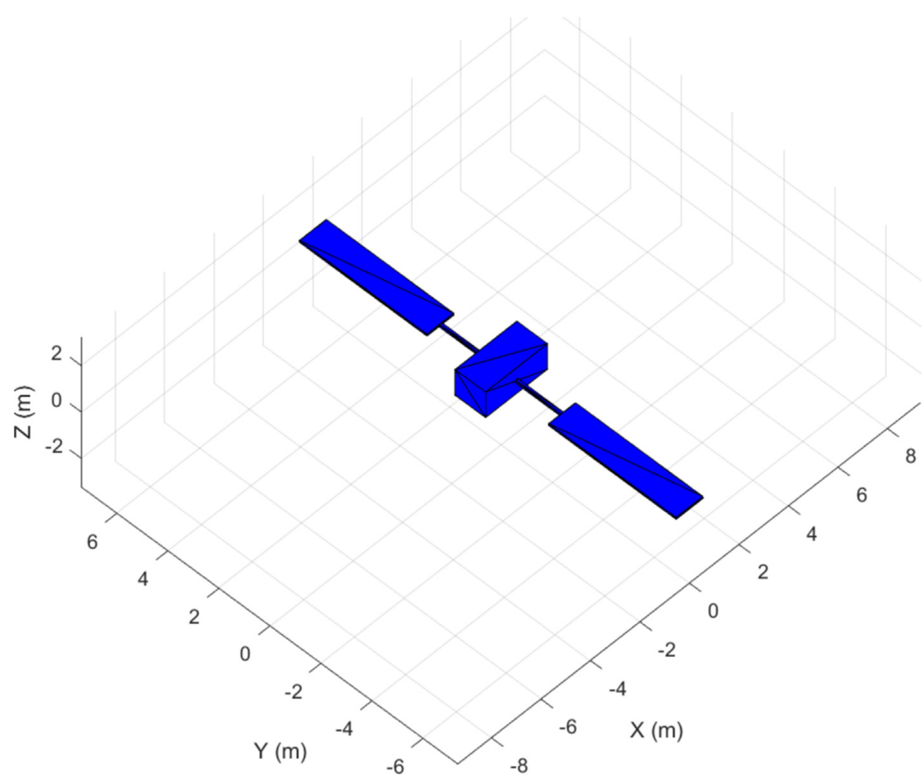

Fig. 12. Representation of the box-wing model.

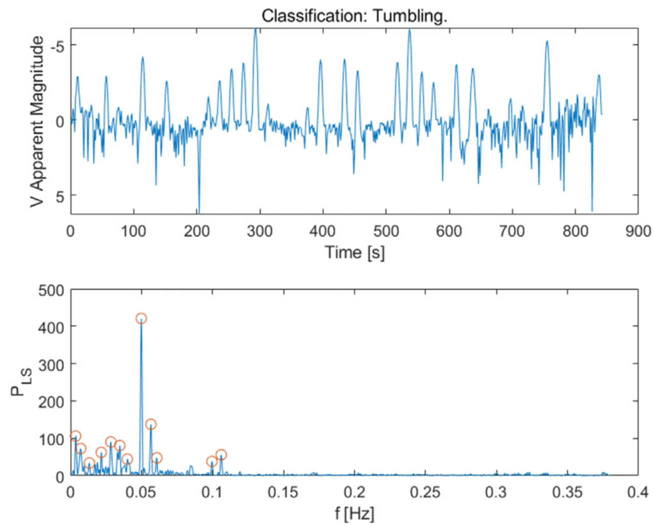

Fig. 13. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a correct classification for the first tumbling scenario, considering $[\psi_0, \theta_0, \varphi_0]$ = [0°, 0°, 0°] and ω_0 = [0°/s, −3°/s, 4°/s].

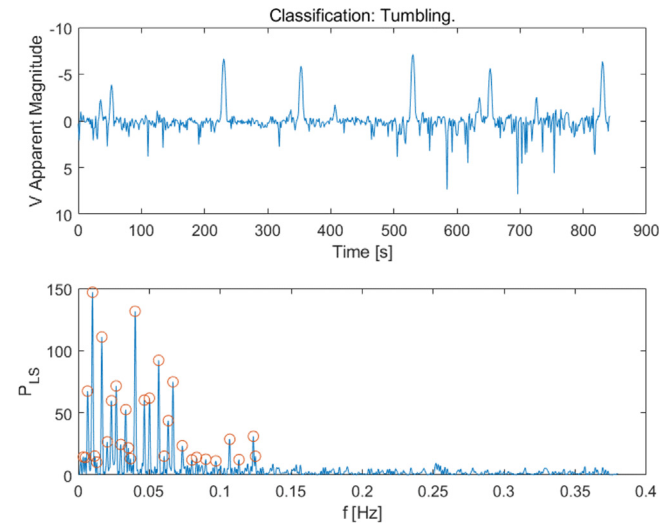

Fig. 15. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a correct classification for the first tumbling scenario, considering $[\psi_0, \theta_0, \varphi_0]$ = [-113°, 57°, −165°] and ω_0 = [0°/s, 1°/s, 6°/s].

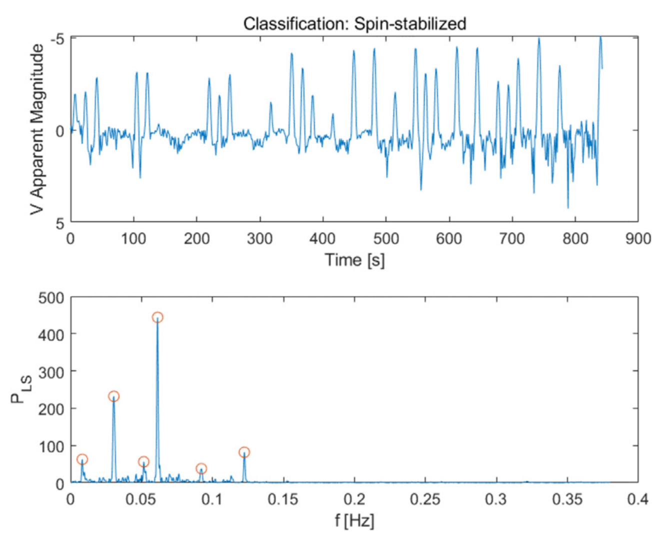

Fig. 14. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a wrong classification for the first tumbling scenario, considering $[\psi_0, \theta_0, \varphi_0]$ = [0°, 0°, 0°] and ω_0 = [-2°/s, −2°/s, 5°/s].

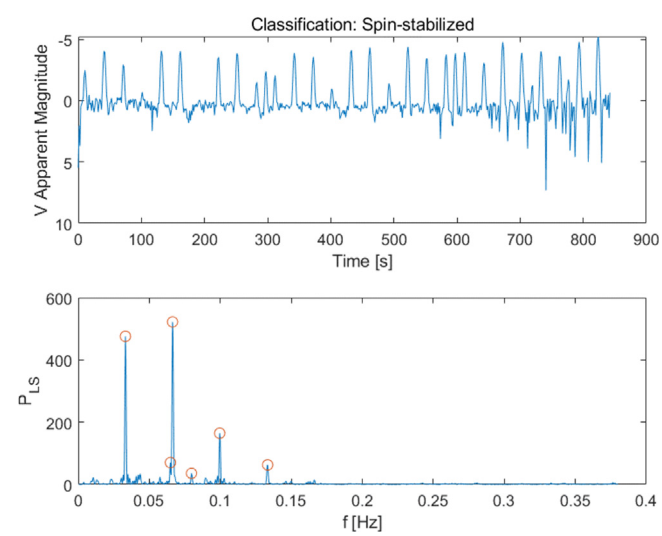

Fig. 16. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a wrong classification for the first tumbling scenario, considering $[\psi_0, \theta_0, \varphi_0]$ = [-66°, 24°, −151°] and ω_0 = [0°/s, 1°/s, 6°/s].

attitude equal to $[\psi_0, \theta_0, \varphi_0]$ = [-113°, 57°, −165°], showing the high number of peaks in the LSP, i.e., much higher than the threshold on the number of peaks set to 7. An example of wrong spin-stabilized classification is also reported in Fig. 16. Here, the frequencies of the first and third peaks in the periodogram are multiples and equal to 0.0332 Hz and 0.0664 Hz, respectively, with the corresponding rotation periods resulting to be among the best 5 ones minimizing ρ in the PDM analysis.

Second tumbling scenario

To assess the impact of the object's model on the tumbling classification performance, the rocket body model presented previously in this section (Fig. 5 and Table 1) has been considered, repeating the simulations using the same set of initial attitude parameters of the second analysis of the first tumbling scenario. The results have shown that the number of misclassifications increased a lot, with 174 out of 500 cases (34.8 %) wrongly classified as "Spin-stabilized", 319 correct classifications (63.8 %) and 7 cases marked as "Uncertain" (1.4 %). Fig. 17 shows the simulated light curve and the periodogram of the wrong classification obtained considering the same simulated tumbling conditions as in Fig. 15, which corresponded to a correct classification using the Galileo box-wing model. Here, the light curve appears to be very regular and the periodogram

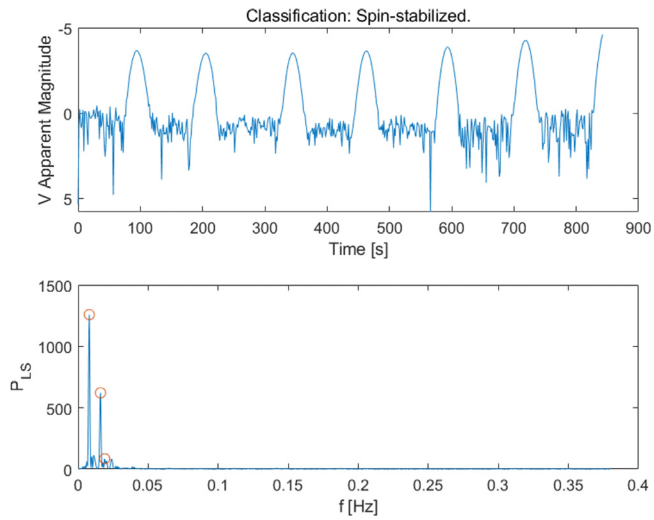

Fig. 17. Simulated light curve after detrending (top) and corresponding LSP (bottom) of a wrong classification for the second tumbling scenario, considering $[\psi_0, \theta_0, \varphi_0]$ = [-113°, 57°, −165°] and ω_0 = [0°/s, 1°/s, 6°/s].

is characterized by three peaks, the first two occurring at multiple frequencies. The degradation of the performance can be ascribed to the shape of the target object used in this second scenario, i.e., the rocket body. Indeed, the observed brightness of a cylinder-shape object can exhibit a periodic pattern regardless of the actual attitude, due to its symmetry as seen by the ground sensor.

5. Real test cases results

In this section, the classification algorithm is applied to several light curves extracted from the Mini-Mega Tortora (MMT) database, in which more than 470,000 tracks of around 13,800 space objects, collected over a time span of almost 10 years from June 2014 to date, are publicly available (Karpov et al., 2016). The MMT system is a multi-purpose wide-field monitoring instrument owned by the Kazan Federal University and operated together with the Special Astrophysical Observatory in Russia, including a database of meteor tracks and a database of tracks of satellites observed by MMT and identified using public TLE data (Beskin et al., 2017). The geodetic coordinates of the MMT instrument are 43.650°N latitude, 41.431°E longitude, 2.030 km altitude. For each track, the MMT database reports the visual magnitude and a rough classification of its variability. Indeed, three kinds of light curve classifications are considered, depending on the periodicity: periodic, when the light curve presents a clear pattern and a rotation period can be computed; aperiodic, when the light curve presents stochastic brightness variations or the data is not enough to estimate a period; non-variable, when the brightness is not varying significantly.

The performance of the light curve classifier has been tested by applying the algorithm to light curves of three different RSOs: two rocket bodies classified as periodic by the MMT database, with quite different rotation periods, and a satellite classified as aperiodic. Clearly, since the rough classifications performed by the MMT database refers to the single tracks, also objects defined as periodic can present light curves classified as non-variable or aperiodic. Some of these tracks have been tested too, to verify the results of the proposed algorithm in these situations. The main characteristics of the three satellites have been retrieved by MMT database and Celestrak website ("CelesTrak," n.d.) and are summarized in Table 5, in which also the variability class of the satellites is reported.

5.1. Minotaur R/B (periodic)

The Minotaur R/B is a vehicle launched on February 2011 by the United States (US). Most of the tracks are in the time interval from 2014 to 2017, showing a change in the rotation period, which increases from around 38 s estimated in 2014 to around 65 s in 2017. In the following, the rotation periods reported by the MMT database will be referred as "true" periods, while the periods estimated by the proposed classification algorithm will be referred as "estimated" periods. Five periodic tracks of the Minotaur R/B have been considered for the analysis (Table 6), four of which characterized by a true period lower than 50 s, and the last one by a period of 65 s. While the classifier shows very good results in the first four cases, it fails in estimating the right rotation period in the last case, despite a correct "Spin-stabilized" classification. Indeed, the root mean square error, RMSE, of the estimated periods is equal to 14.806 s when considering all the tracks, while reducing to 0.684 s when excluding the last track from the statistics. The reason for this wrong period estimation is that the true period of the last case is more than half the duration of the track: this represents an issue for the proposed methodology, since the candidate periods for the PDM are sampled in a time interval whose upper bound is half the track duration, to ensure the pattern is repeated at least two times when folding the curve (as explained in Section 2). However, it can be noted that the estimated period is almost half the true period, thus suggesting that the algorithm recognizes a pattern in the light curves, but data is not enough to compute the right spin period. Fig. 18, instead, shows the light curve (after detrending) and the corresponding LSP for the first case of Table 6, in which the periodic behaviour is identified and the rotation period is accurately estimated.

Table 5
Main characteristics of the three satellites considered for the analysis.

	MINOTAUR R/B	N-1 R/B	ENVISAT
NORAD ID	37365	10675	27386
Orbital period (min)	109.77	107.08	100.11
Inclination (°)	90.15	69.36	98.24
Apogee height (km)	1230	1209	771.5
Perigee height (km)	1203	975	769.9
Variability class	Periodic	Periodic	Aperiodic

Table 6
Results of the analysis for the Minotaur R/B. Five periodic tracks have been considered.

Track	UTC epoch	Track duration (s)	True period (s)	Classification	Estimated period (s)
1	01/09/2014 01:27:39	142.7	38.19	Spin-stabilized	38.07
2	13/09/2014 00:54:39	127.4	37.07	Spin-stabilized	37.78
3	24/02/2015 01:15:22	130.4	44.13	Spin-stabilized	45.29
4	09/03/2015 00:27:37	135.9	45.22	Spin-stabilized	45.29
5	15/09/2017 01:09:51	124.5	65.20	Spin-stabilized	33.51

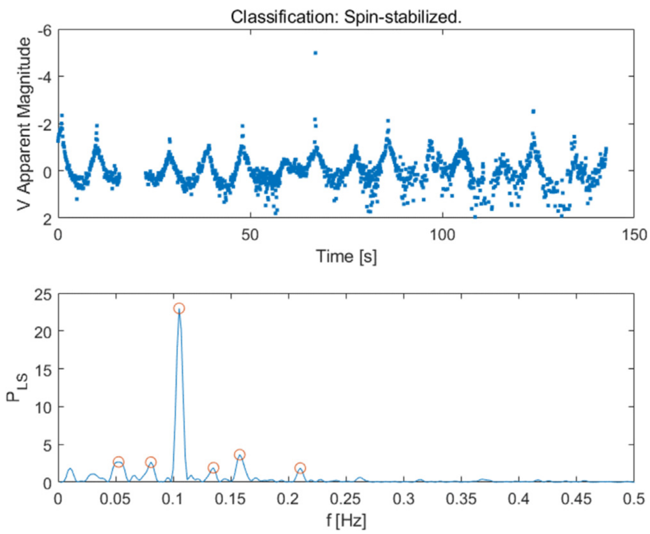

Fig. 18. Real light curve after detrending (top) and corresponding LSP (bottom) of a correct classification for the first test case of the Minotaur R/B analysis.

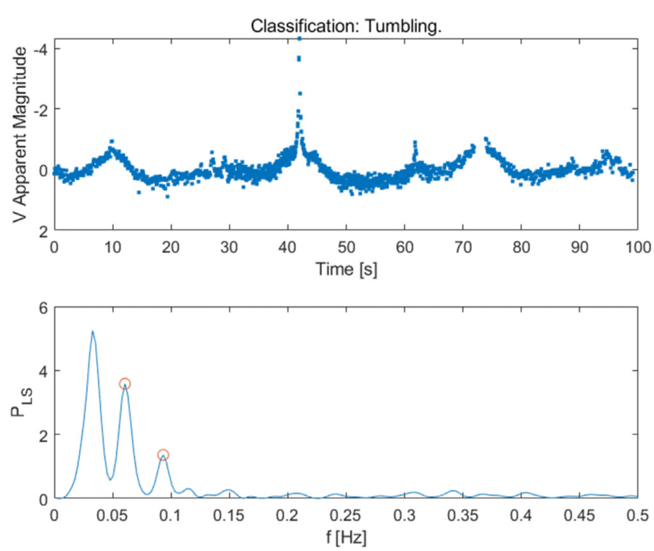

Fig. 19. Real light curve after detrending (top) and corresponding LSP (bottom) of a correct classification for an aperiodic track of the Minotaur R/B, correctly classified as Tumbling by the algorithm.

An example of an aperiodic track of Minotaur R/B is also shown in Fig. 19, with the corresponding periodogram. The track starts on 05/06/2017 18:33:51 UTC and lasts 99.2 s. The light curve appears very different from the one of Fig. 18, making it difficult to recognize a pattern. The algorithm correctly classifies the track as tumbling, with two peaks at non-multiple frequencies found in the LSP. It is worth noting that the highest value of the LSP, which should have been the first peak, is discarded, being lower than 2 times the frequency corresponding to $\frac{1}{T_{obs}}$, as described in Section 2.

5.2. N-1 R/B (periodic)

The second rocket body considered is N-1, a Japanese vehicle launched in February 1978. Many periodic tracks are available between 2014 and 2023, with rotation periods ranging from 5 s to 7 s. Here, 10 different periodic tracks have been selected and the results of the classification are shown in Table 7. The classifier always performs well, also accurately estimating the rotation period, i.e., RMSE equal to 0.0311 s for the correct classifications, except for one case in which the classification is Uncertain. This missed classification is probably due to the presence of many gaps in the final part of light curve, quite evident in Fig. 20. A further test has been done reducing the observation windows to the first 60 s to neglect the gaps in the light curve and the result is shown in Fig. 21. By doing so, the algorithm correctly classifies the track and the estimated rotation period results to be 6.55 s, very close to the true one. Finally, the result of the last N-1 R/B test case is plotted in Fig. 22 as an example, showing the good behaviour of the classifier even dealing with a very short light curve, with a track duration of only 36.8 s. Indeed, differently for last test case of the Minotaur analysis, the rotation period is much lower than half the width of the observation window, so the algorithm succeeds in correctly classifying the track and estimating the right spin period.

5.3. Envisat (aperiodic)

ENVISAT (Environmental satellite) was an Earth-observing satellite, now inactive, launched on March 2002 and operated by European Space Agency (ESA). To date Envisat results to be uncontrolled and slowly tumbling (Sagnières and Sharf, 2018; Sommer et al., 2017). Indeed, it is classified as an aperiodic object within the MMT database, since almost all the tracks are classified as aperiodic, with a few light curves tagged as “Non-variable”. The

Table 7
Results of the analysis for the N-1 R/B. Ten periodic tracks have been considered.

Track	UTC epoch	Track duration (s)	True period (s)	Classification	Estimated period (s)
1	24/08/2014 19:02:29	91.1	5.20	Spin-stabilized	5.20
2	18/10/2015 15:53:35	70.5	5.41	Spin-stabilized	5.43
3	11/08/2016 19:01:06	68.9	5.71	Spin-stabilized	5.68
4	08/03/2017 01:46:56	103.4	5.78	Spin-stabilized	5.74
5	20/08/2018 00:31:16	143.2	6.13	Spin-stabilized	6.09
6	10/05/2019 18:59:31	83.7	6.38	Spin-stabilized	6.33
7	01/07/2020 23:12:08	105.7	6.55	Uncertain	–
7 (shortened)	01/07/2020 23:12:08	66.1	6.55	Spin-stabilized	6.54
8	12/10/2021 00:37:29	73.9	6.97	Spin-stabilized	6.96
9	21/04/2022 19:13:30	92.7	7.13	Spin-stabilized	7.13
10	29/12/2023 15:26:37	36.8	7.72	Spin-stabilized	7.76

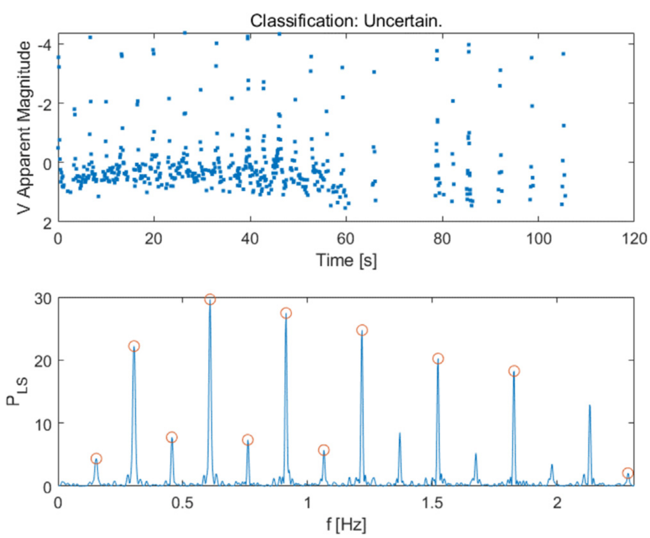

Fig. 20. Real light curve after detrending (top) and corresponding LSP (bottom) of a missed classification for the seventh test case of the N-1 R/B analysis.

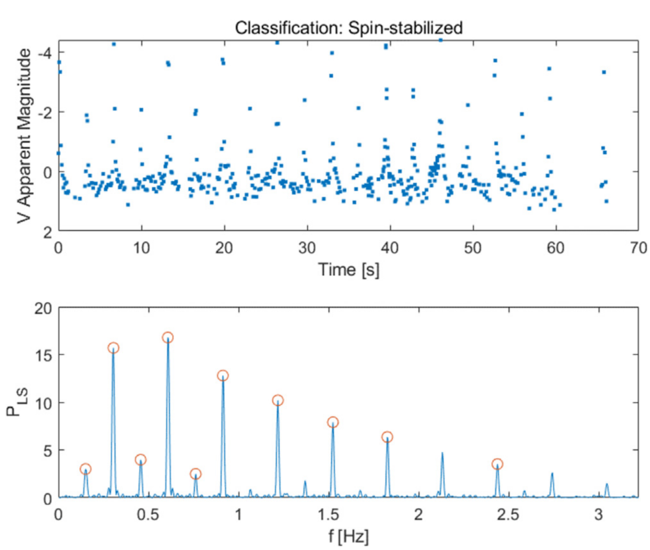

Fig. 21. Real light curve after detrending (top) and corresponding LSP (bottom) of the seventh test case of the N-1 R/B analysis, reducing the observation window to remove the gaps, resulting in a correct classification.

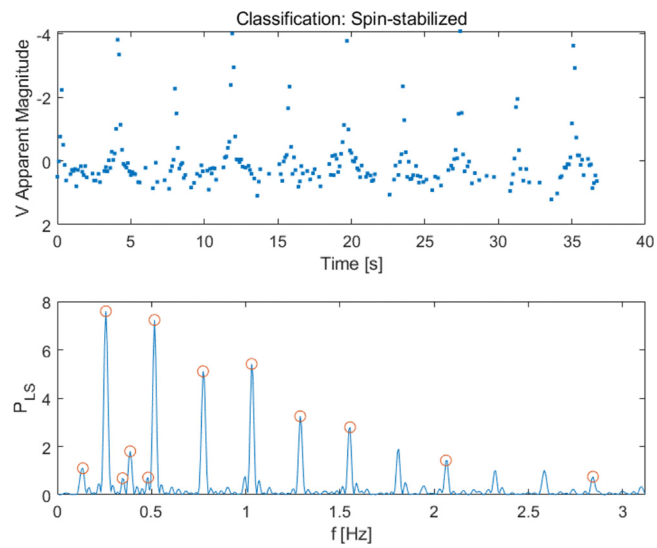

Fig. 22. Real light curve after detrending (top) and corresponding LSP (bottom) of a correct classification for the last test case of the N-1 R/B analysis.

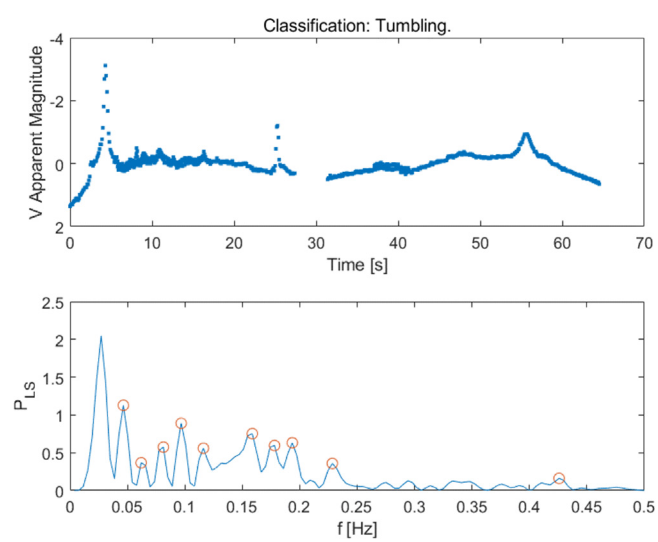

Fig. 23. Real light curve after detrending (top) and corresponding LSP (bottom) of a correct tumbling classification for the third test case of the Envisat analysis.

Table 8
Results of the analysis for the Envisat satellite. Five aperiodic tracks and one non-variable track have been considered.

Track	UTC epoch	Track duration (s)	MMT classification	Classification
1	31/07/2015 18:30:53	61.7	Aperiodic	Tumbling
2	02/09/2017 16:55:33	38.3	Aperiodic	Tumbling
3	22/10/2021 02:18:32	64.4	Aperiodic	Tumbling
4	31/08/2022 01:17:26	66.8	Aperiodic	Tumbling
5	19/08/2023 01:04:19	94.2	Aperiodic	Tumbling
6	28/09/2023 01:02:18	66.7	Non variable	Tumbling

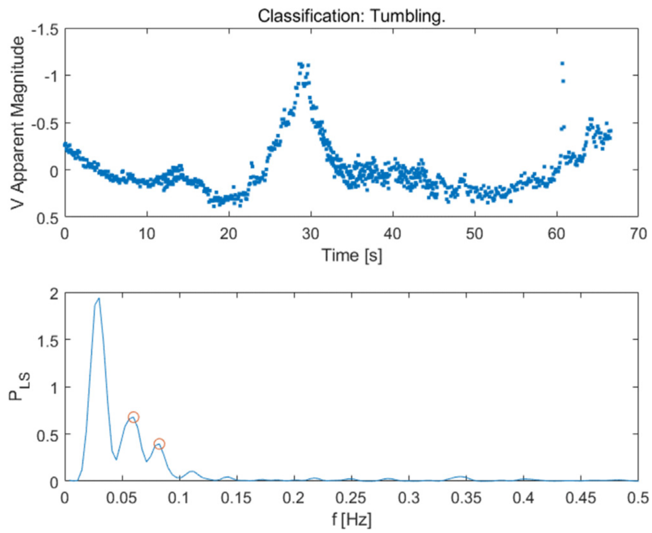

Fig. 24. Real light curve after detrending (top) and corresponding LSP (bottom) of a tumbling classification for the sixth test case of the Envisat analysis. The track is tagged as "Non-variable" in the MMT database.

results of the classification for five aperiodic tracks and one non-variable track are presented in Table 8.

The algorithm has shown good performance, always classifying the aperiodic tracks as Tumbling. The light curve and the LSP for the third test case are depicted in Fig. 23, with the LSP presenting 10 peaks at non-multiple frequencies, thus resulting in the tumbling classification. Fig. 24, instead, shows the results for the last track classified as non-variable by MMT database. Also here, the proposed classification algorithm tagged the track as "Tumbling", finding two peaks in the LSP at non-multiple frequencies. Moreover, as already noted in Fig. 19 for an aperiodic Minotaur R/B test case, the highest value of the LSP is discarded for the same reason.

To sum up, the classification algorithm has proven to be quite reliable in classifying the real light curves retrieved by the MMT database, both periodic and aperiodic. In particular, for periodic tracks, also the rotation period estimation resulted to be accurate, with values of RMSE of the periods of correctly classified tracks with respect to those reported on MMT database equal to 0.6836 and 0.0311 for the Minotaur and N-1 test cases, respectively. The correct estimation of rotation periods has proved to be valid as long as the time length of the observation windows is at least two times the spin period of the objects. This aspect can be investigated more in depth, analyzing the possibility to extend the time interval in which the candidate periods required by the PDM technique are sampled, trying to reconstruct also greater spin periods.

6. Conclusion

In this paper, an architecture based on direct analysis of light curves is proposed to address the problem of classification of RSOs' attitude motion when no a-priori information of the target object is available. The architecture exploits the Lomb-Scargle Periodogram and the Phase Dispersion Minimization algorithms to classify the object's attitude motion as "Spin-stabilized", "Tumbling" or "Uncertain".

First, the classification algorithm has been tested for synthetically generated light curves corresponding to both spin-stabilized and tumbling test cases. For the spin-stabilized scenarios, the results have shown that the percentage of correct classification is more than 90 % for angular velocities between 3°/s and 30°/s, also for short observation windows, while missed or wrong classification can be related to the presence of more irregular regions as well as zones characterized by lower oscillation amplitude within the light curve. For the tumbling test cases, a classification accuracy of around 90 % is achieved for different sets of attitude states when considering a typical box-wing configuration for the simulated target object, while it is more likely that a wrong "Spin-stabilized" classification can occur when observing a cylinder-shape object, which can exhibit a periodic pattern in the light curve regardless of the actual attitude motion.

The classifier has also been applied to several test cases retrieved by the MMT database, analyzing two spinning rocket bodies and a tumbling inactive satellite, labelled as periodic or aperiodic in the exploited public database, respectively. The algorithm has shown excellent performance in classifying the real light curves and, for periodic tracks, also the rotation period estimation resulted to be very accurate. Indeed, the values of RMSE of the rotation periods with respect to those reported on MMT database are always lower than 1 s, as long as the time length of the observation windows is at least two times the spin period of the objects, since the current PDM implementation only exploits half of the track duration.

In this perspective, an update to the classification algorithm can regard the application of the PDM technique. Indeed, since the PDM is currently based on the convolution of the whole light curve with a candidate period, this approach is expected to be less effective in case of highly variable curves, which can often be the case for RSOs, also over a single observation window. Therefore, the possibility to use the PDM in combination with a sliding window over the observation time interval could be investigated, thus better exploiting the characteristics of the PDM technique.

Moreover, a more comprehensive analysis of the impact of user-defined parameters on the performance of the algorithm should be carried out. Finally, the possibility to take multi-temporal and multi-sensor measurements into account to increase the reliability of the classification can be investigated, in order to improve the overall performance of the algorithm, thus enhancing its capability to support civilian and military operational applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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